NEGATIVE VANISHING AND LEFT EXACTNESS OF THE ZEROTH RIGHT DERIVED FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5795-5833, following the statement at 5781-5793.

Editorial note. No omissions: all three parts executed; the cofinality reduction to tau_{>=0} replacements is argued in full via two Categories lemmas.

Allowed standard prerequisites (editorial).

- Vanishing of RF in negative degrees. lemma-negative-vanishing (same chapter, cited around line 5797). Gives R^iF(A[0]) = 0 for i < 0, used in parts (1) and (2).
- Canonical triangle of a short exact sequence. lemma-derived-canonical-delta-functor (same chapter, cited around line 5806). Produces the distinguished triangle (A[0], B[0], C[0]) in D^+(A).
- Essentially constant systems and cofinal restriction. Categories, Lemma categories-lemma-image-essentially-constant and categories-lemma-cofinal-essentially-constant (cross-chapter, cited around lines 5821 and 5826). Identifies R^0F(A) with H^0 of the essentially constant system and justifies restricting the index system to complexes concentrated in nonnegative degrees.

Source reference. Original reference 0A1S: focused-seeds/geometry/S9-der-left-exact-higher-derived/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05SH: focused-seeds/geometry/S9-der-left-exact-higher-derived/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0152: sources/stacks-derived/pinned-original-source.tex, source lines 4182–4182. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TC: sources/stacks-derived/pinned-original-source.tex, source lines 5714–5714. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories and assume $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is everywhere defined.*

(1) *We have $R^iF = 0$ for $i < 0$,*
(2) *R^0F is left exact,*
(3) *the map $F \to R^0F$ is an isomorphism if and only if F is left exact.*

Proof. Let A be an object of $\mathcal{A}$. By Lemma 05TC we have $H^i(RF(A[0]) = 0$ for $i < 0$. This proves (1).

Let $0 \to A \to B \to C \to 0$ be a short exact sequence of $\mathcal{A}$. By Lemma 0152 we obtain a distinguished triangle $(A[0], B[0], C[0], a, b, c)$ in $D^+(\mathcal{A})$. From the long exact cohomology sequence (and the vanishing for $i < 0$ proved above) we deduce that $0 \to R^0F(A) \to R^0F(B) \to R^0F(C)$ is exact. Hence R^0F is left exact. Of course this also proves that if $F \to R^0F$ is an isomorphism, then F is left exact.

Assume F is left exact. Recall that $RF(A[0])$ is the value of the essentially constant system $F(K^\bullet)$ for $s : A[0] \to K^\bullet$ quasi-isomorphisms. It follows that $R^0F(A)$ is the value of the essentially constant system $H^0(F(K^\bullet))$ for $s : A[0] \to K^\bullet$ quasi-isomorphisms, see Categories, Lemma 05SH. But if $s : A[0] \to K^\bullet$ is a quasi-isomorphism, then $A[0] \to \tau_{\geq 0}K^\bullet$ is a quasi-isomorphism. Hence in the category $A[0]/\mathrm{Qis}^+(\mathcal{A})$ the quasi-isomorphisms $s : A[0] \to K^\bullet$ with $K^n = 0$ for $n < 0$ are cofinal. It follows from Categories, Lemma 0A1S that we may restrict to such s. Moreover, for such an s the sequence

$$0 \to A \to K^0 \to K^1$$

is exact. Since F is left exact we see that $0 \to F(A) \to F(K^0) \to F(K^1)$ is exact as well. It follows that $F(A) \to H^0(F(K^\bullet))$ is an isomorphism and the system is actually constant with value $F(A)$. We conclude $R^0F = F$ as desired. □